



MATH230: Vectors in 3D

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Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Dot product

1. Find the dot product $u \cdot v$, where $u = (-3, -4, 0)$ and $v = (-1, -1, -2)$.

$$(-3, -4, 0) \cdot (-1, -1, -2)$$

Answer: _____

2. Force vector $F = (1, 3, 6)$ N acts on a displacement $d = (2, 4, 3)$ m. Compute the work $W = F \cdot d$.

$$(1, 3, 6) \cdot (2, 4, 3)$$

Answer: _____

3. Compute the dot product: $(0, 1, 1) \cdot (2, 3, -3)$. Are these vectors orthogonal?

$$(0, 1, 1) \cdot (2, 3, -3)$$

Answer: _____

4. Find the dot product $u \cdot v$, where $u = (-3, 2, -4)$ and $v = (-1, 4, -2)$.

$$(-3, 2, -4) \cdot (-1, 4, -2)$$

Answer: _____

5. Force vector $F = (3, 1, 4)$ N acts on a displacement $d = (5, 4, 5)$ m. Compute the work $W = F \cdot d$.

$$(3, 1, 4) \cdot (5, 4, 5)$$

Answer: _____

6. Compute the dot product: $(1, -1, -2) \cdot (1, 2, 1)$. Are these vectors orthogonal?

$$(1, -1, -2) \cdot (1, 2, 1)$$

Answer: _____



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7. Find the dot product $u \cdot v$, where $u = (3, 0, 1)$ and $v = (-1, 3, 1)$.

$$(3, 0, 1) \cdot (-1, 3, 1)$$

Answer: _____

8. Force vector $F = (2, 5, 5)$ N acts on a displacement $d = (2, 2, 2)$ m. Compute the work $W = F \cdot d$.

$$(2, 5, 5) \cdot (2, 2, 2)$$

Answer: _____

9. Compute the dot product: $(-1, -3, 2) \cdot (1, 0, 3)$. Are these vectors orthogonal?

$$(-1, -3, 2) \cdot (1, 0, 3)$$

Answer: _____

10. Find the dot product $u \cdot v$, where $u = (-2, 3, 4)$ and $v = (0, 3, 3)$.

$$(-2, 3, 4) \cdot (0, 3, 3)$$

Answer: _____

11. Force vector $F = (1, 2, 2)$ N acts on a displacement $d = (3, 1, 2)$ m. Compute the work $W = F \cdot d$.

$$(1, 2, 2) \cdot (3, 1, 2)$$

Answer: _____

12. Compute the dot product: $(3, 1, -3) \cdot (0, 2, 1)$. Are these vectors orthogonal?

$$(3, 1, -3) \cdot (0, 2, 1)$$

Answer: _____

13. Find the dot product $u \cdot v$, where $u = (-1, -1, -1)$ and $v = (-4, -1, 3)$.

$$(-1, -1, -1) \cdot (-4, -1, 3)$$

Answer: _____

14. Force vector $F = (2, 1, 6)$ N acts on a displacement $d = (4, 1, 2)$ m. Compute the work $W = F \cdot d$.

$$(2, 1, 6) \cdot (4, 1, 2)$$

Answer: _____

15. Compute the dot product: $(-1, -1, 1) \cdot (-2, -1, -2)$. Are these vectors orthogonal?

$$(-1, -1, 1) \cdot (-2, -1, -2)$$

Answer: _____



16. Find the dot product $u \cdot v$, where $u = (-2, -3, -1)$ and $v = (3, 0, 1)$.

$$(-2, -3, -1) \cdot (3, 0, 1)$$

Answer: _____

17. Force vector $F = (4, 4, 3)$ N acts on a displacement $d = (3, 4, 1)$ m. Compute the work $W = F \cdot d$.

$$(4, 4, 3) \cdot (3, 4, 1)$$

Answer: _____

18. Compute the dot product: $(0, -3, 2) \cdot (1, 0, 1)$. Are these vectors orthogonal?

$$(0, -3, 2) \cdot (1, 0, 1)$$

Answer: _____

19. Find the dot product $u \cdot v$, where $u = (-3, 0, 2)$ and $v = (-1, 3, 0)$.

$$(-3, 0, 2) \cdot (-1, 3, 0)$$

Answer: _____

20. Force vector $F = (1, 1, 4)$ N acts on a displacement $d = (4, 4, 3)$ m. Compute the work $W = F \cdot d$.

$$(1, 1, 4) \cdot (4, 4, 3)$$

Answer: _____

21. Compute the dot product: $(2, 0, 2) \cdot (2, 3, 1)$. Are these vectors orthogonal?

$$(2, 0, 2) \cdot (2, 3, 1)$$

Answer: _____

22. Find the dot product $u \cdot v$, where $u = (-3, 0, 3)$ and $v = (3, -2, 1)$.

$$(-3, 0, 3) \cdot (3, -2, 1)$$

Answer: _____

23. Force vector $F = (3, 5, 5)$ N acts on a displacement $d = (1, 3, 2)$ m. Compute the work $W = F \cdot d$.

$$(3, 5, 5) \cdot (1, 3, 2)$$

Answer: _____

Vector magnitude

24. Find the magnitude of vector $v = (2, 3, 3)$.

$$|(2, 3, 3)|$$

Answer: _____



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25. Find the magnitude of vector $v = (0, 2, 3)$.

$$|(0, 2, 3)|$$

Answer: _____

26. Find the magnitude of vector $v = (4, 1, 3)$.

$$|(4, 1, 3)|$$

Answer: _____

27. Find the magnitude of vector $v = (0, 2, 0)$.

$$|(0, 2, 0)|$$

Answer: _____

28. Find the magnitude of vector $v = (3, 1, 1)$.

$$|(3, 1, 1)|$$

Answer: _____

29. Find the magnitude of vector $v = (3, 2, 2)$.

$$|(3, 2, 2)|$$

Answer: _____

30. Find the magnitude of vector $v = (0, 0, 1)$.

$$|(0, 0, 1)|$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Topics: Vector magnitude, Dot product. All answers verified by independent computation.

Solutions



Dot product

1. Find the dot product $u \cdot v$, where $u = (-3, -4, 0)$ and $v = (-1, -1, -2)$.

$$(-3, -4, 0) \cdot (-1, -1, -2)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-3)(-1) + (-4)(-1) + (0)(-2) = 7.$$

Answer: $= 7$

2. Force vector $F = (1, 3, 6)$ N acts on a displacement $d = (2, 4, 3)$ m. Compute the work $W = F \cdot d$.

$$(1, 3, 6) \cdot (2, 4, 3)$$

$$\rightarrow W = F \cdot d = (1)(2) + (3)(4) + (6)(3) = 32 \text{ J.}$$

Answer: $= 32$

3. Compute the dot product: $(0, 1, 1) \cdot (2, 3, -3)$. Are these vectors orthogonal?

$$(0, 1, 1) \cdot (2, 3, -3)$$

$$\rightarrow \text{Dot product} = (0)(2) + (1)(3) + (1)(-3) = 0.$$

$$\rightarrow \text{If dot product} = 0 \text{ the vectors are orthogonal; otherwise they are not.}$$

Answer: $= 0$

4. Find the dot product $u \cdot v$, where $u = (-3, 2, -4)$ and $v = (-1, 4, -2)$.

$$(-3, 2, -4) \cdot (-1, 4, -2)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-3)(-1) + (2)(4) + (-4)(-2) = 19.$$

Answer: $= 19$

5. Force vector $F = (3, 1, 4)$ N acts on a displacement $d = (5, 4, 5)$ m. Compute the work $W = F \cdot d$.

$$(3, 1, 4) \cdot (5, 4, 5)$$

$$\rightarrow W = F \cdot d = (3)(5) + (1)(4) + (4)(5) = 39 \text{ J.}$$

Answer: $= 39$

6. Compute the dot product: $(1, -1, -2) \cdot (1, 2, 1)$. Are these vectors orthogonal?

$$(1, -1, -2) \cdot (1, 2, 1)$$

$$\rightarrow \text{Dot product} = (1)(1) + (-1)(2) + (-2)(1) = -3.$$

$$\rightarrow \text{If dot product} = 0 \text{ the vectors are orthogonal; otherwise they are not.}$$

Answer: $= -3$

7. Find the dot product $u \cdot v$, where $u = (3, 0, 1)$ and $v = (-1, 3, 1)$.

$$(3, 0, 1) \cdot (-1, 3, 1)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (3)(-1) + (0)(3) + (1)(1) = -2.$$

Answer: $= -2$



8. Force vector $F = (2, 5, 5)$ N acts on a displacement $d = (2, 2, 2)$ m. Compute the work $W = F \cdot d$.

$$(2, 5, 5) \cdot (2, 2, 2)$$

$$\rightarrow W = F \cdot d = (2)(2) + (5)(2) + (5)(2) = 24 \text{ J.}$$

Answer: $= 24$

9. Compute the dot product: $(-1, -3, 2) \cdot (1, 0, 3)$. Are these vectors orthogonal?

$$(-1, -3, 2) \cdot (1, 0, 3)$$

$$\rightarrow \text{Dot product} = (-1)(1) + (-3)(0) + (2)(3) = 5.$$

$\rightarrow$ If dot product = 0 the vectors are orthogonal; otherwise they are not.

Answer: $= 5$

10. Find the dot product $u \cdot v$, where $u = (-2, 3, 4)$ and $v = (0, 3, 3)$.

$$(-2, 3, 4) \cdot (0, 3, 3)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-2)(0) + (3)(3) + (4)(3) = 21.$$

Answer: $= 21$

11. Force vector $F = (1, 2, 2)$ N acts on a displacement $d = (3, 1, 2)$ m. Compute the work $W = F \cdot d$.

$$(1, 2, 2) \cdot (3, 1, 2)$$

$$\rightarrow W = F \cdot d = (1)(3) + (2)(1) + (2)(2) = 9 \text{ J.}$$

Answer: $= 9$

12. Compute the dot product: $(3, 1, -3) \cdot (0, 2, 1)$. Are these vectors orthogonal?

$$(3, 1, -3) \cdot (0, 2, 1)$$

$$\rightarrow \text{Dot product} = (3)(0) + (1)(2) + (-3)(1) = -1.$$

$\rightarrow$ If dot product = 0 the vectors are orthogonal; otherwise they are not.

Answer: $= -1$

13. Find the dot product $u \cdot v$, where $u = (-1, -1, -1)$ and $v = (-4, -1, 3)$.

$$(-1, -1, -1) \cdot (-4, -1, 3)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-1)(-4) + (-1)(-1) + (-1)(3) = 2.$$

Answer: $= 2$

14. Force vector $F = (2, 1, 6)$ N acts on a displacement $d = (4, 1, 2)$ m. Compute the work $W = F \cdot d$.

$$(2, 1, 6) \cdot (4, 1, 2)$$

$$\rightarrow W = F \cdot d = (2)(4) + (1)(1) + (6)(2) = 21 \text{ J.}$$

Answer: $= 21$



15. Compute the dot product: $(-1, -1, 1) \cdot (-2, -1, -2)$. Are these vectors orthogonal?

$$(-1, -1, 1) \cdot (-2, -1, -2)$$

$$\rightarrow \text{Dot product} = (-1)(-2) + (-1)(-1) + (1)(-2) = 1.$$

$\rightarrow$ If dot product = 0 the vectors are orthogonal; otherwise they are not.

Answer: = 1

16. Find the dot product $u \cdot v$, where $u = (-2, -3, -1)$ and $v = (3, 0, 1)$.

$$(-2, -3, -1) \cdot (3, 0, 1)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-2)(3) + (-3)(0) + (-1)(1) = -7.$$

Answer: = - 7

17. Force vector $F = (4, 4, 3)$ N acts on a displacement $d = (3, 4, 1)$ m. Compute the work $W = F \cdot d$.

$$(4, 4, 3) \cdot (3, 4, 1)$$

$$\rightarrow W = F \cdot d = (4)(3) + (4)(4) + (3)(1) = 31 \text{ J.}$$

Answer: = 31

18. Compute the dot product: $(0, -3, 2) \cdot (1, 0, 1)$. Are these vectors orthogonal?

$$(0, -3, 2) \cdot (1, 0, 1)$$

$$\rightarrow \text{Dot product} = (0)(1) + (-3)(0) + (2)(1) = 2.$$

$\rightarrow$ If dot product = 0 the vectors are orthogonal; otherwise they are not.

Answer: = 2

19. Find the dot product $u \cdot v$, where $u = (-3, 0, 2)$ and $v = (-1, 3, 0)$.

$$(-3, 0, 2) \cdot (-1, 3, 0)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-3)(-1) + (0)(3) + (2)(0) = 3.$$

Answer: = 3

20. Force vector $F = (1, 1, 4)$ N acts on a displacement $d = (4, 4, 3)$ m. Compute the work $W = F \cdot d$.

$$(1, 1, 4) \cdot (4, 4, 3)$$

$$\rightarrow W = F \cdot d = (1)(4) + (1)(4) + (4)(3) = 20 \text{ J.}$$

Answer: = 20

21. Compute the dot product: $(2, 0, 2) \cdot (2, 3, 1)$. Are these vectors orthogonal?

$$(2, 0, 2) \cdot (2, 3, 1)$$

$$\rightarrow \text{Dot product} = (2)(2) + (0)(3) + (2)(1) = 6.$$

$\rightarrow$ If dot product = 0 the vectors are orthogonal; otherwise they are not.

Answer: = 6



22. Find the dot product $u \cdot v$, where $u = (-3, 0, 3)$ and $v = (3, -2, 1)$.

$$(-3, 0, 3) \cdot (3, -2, 1)$$

$$\rightarrow u \cdot v = a_1b_1 + a_2b_2 + a_3b_3.$$

$$\rightarrow = (-3)(3) + (0)(-2) + (3)(1) = -6.$$

Answer: $= -6$

23. Force vector $F = (3, 5, 5)$ N acts on a displacement $d = (1, 3, 2)$ m. Compute the work $W = F \cdot d$.

$$(3, 5, 5) \cdot (1, 3, 2)$$

$$\rightarrow W = F \cdot d = (3)(1) + (5)(3) + (5)(2) = 28 \text{ J}.$$

Answer: $= 28$



Vector magnitude

24. Find the magnitude of vector $v = (2, 3, 3)$.

$$|(2, 3, 3)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{2^2 + 3^2 + 3^2} = \sqrt{22} = \text{None}$.

Answer: $= \sqrt{22} = \text{None}$

25. Find the magnitude of vector $v = (0, 2, 3)$.

$$|(0, 2, 3)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{0^2 + 2^2 + 3^2} = \sqrt{13} = \text{None}$.

Answer: $= \sqrt{13} = \text{None}$

26. Find the magnitude of vector $v = (4, 1, 3)$.

$$|(4, 1, 3)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{4^2 + 1^2 + 3^2} = \sqrt{26} = \text{None}$.

Answer: $= \sqrt{26} = \text{None}$

27. Find the magnitude of vector $v = (0, 2, 0)$.

$$|(0, 2, 0)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{0^2 + 2^2 + 0^2} = \sqrt{4} = 2$.

Answer: $= \sqrt{4} = 2$

28. Find the magnitude of vector $v = (3, 1, 1)$.

$$|(3, 1, 1)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{3^2 + 1^2 + 1^2} = \sqrt{11} = \text{None}$.

Answer: $= \sqrt{11} = \text{None}$

29. Find the magnitude of vector $v = (3, 2, 2)$.

$$|(3, 2, 2)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{3^2 + 2^2 + 2^2} = \sqrt{17} = \text{None}$.

Answer: $= \sqrt{17} = \text{None}$



30. Find the magnitude of vector $v = (0, 0, 1)$.

$$|(0, 0, 1)|$$

→ Formula: $|v| = \sqrt{a^2 + b^2 + c^2}$.

→ $= \sqrt{0^2 + 0^2 + 1^2} = \sqrt{1} = 1$.

Answer: $= \sqrt{1} = 1$

